**MONTHLY ASSESSMENT -III (Detail Design, Prototype and Test phase)**

**(This assessment shall be done by another guide of department or interdepartmentally along with guide)** **(DATE :___/___/___)**

1. Which theoretical subjects/concepts are involved with your project? How it is useful to your project?

   Affective Computing, Machine Learning and Artificial Intelligence, Signal Processing, Human-Computer Interaction (HCI), Psychology and Emotion Theory, Natural Language Processing (NLP), Neuroscience, Music Theory and Musicology, Data Mining and Big Data Analytics, Ethics and Privacy.

2. Which software/design tool/Skills you have learned/applied during the project? Explain the features of it.

   These are just some of the tools and skills that could be applied in developing an emotion-based music player.

   Programming Languages, Machine Learning Libraries, Database Management Systems, User Interface (UI) Design Tools, Version Control Systems, API Integration, Audio Processing Libraries,

   Mobile App Development, Emotion Detection and Analysis.

3. Explain the prototype/model prepared by the student/team.

   Creating a prototype or model for an emotion-based music player involves designing a system that can interpret a user's emotions and tailor music recommendations accordingly.

   Emotion Detection Mechanism

   Music Database

   Testing and Iteration

   Algorithm for Music Selection

4. What are the materials, technology, things have utilized to make the prototype/model?

   Creating a prototype or model for an emotion-based music player involves integrating several technologies and materials. Microcontroller or Single Board Computer, Sensors, EEG Headsets, Cameras, Microphones, Facial Recognition Software, Voice Analysis Software, Music Database, Mobile Applications, Enclosures

5. How many Iterations have you done to reach final solution? Explain modification/revise parameters/characteristics for each iteration.

   Iteration 1: Initial Prototype:: Found that emotion detection accuracy was low.

   Iteration 2: Improved Emotion Detection

   Iteration 3: Expanding Emotion Recognition Modalities

   Iteration 4: Real-time Processing and Feedback

6. What is the scope of the project? How you are planning to implement it in future?

   The scope of an emotion-based music player involves creating a system that can analyze a user's emotional state and select music that aligns with or influences that state. The primary goals are to enhance the user experience by providing personalized music recommendations and to potentially use music as a tool for mood regulation.

**SUGESTIONS BY EVALUATOR:**

| |
|---|
| PROTOTYPE/MODEL SUGGESTIONS: |
| GENERAL SUGESTIONS: |
| Overall Mark, considering assessment I & II (Out of 10): |

**Department/Interdepartmental Evaluator name and sign:**
**Guide sign:**
**Date:**